CS3003: IT Project Management

Dr Georgios Koutsoukos

giorgios.koutsoukos@rhul.ac.uk

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Part 6: Project Controlling/Monitoring

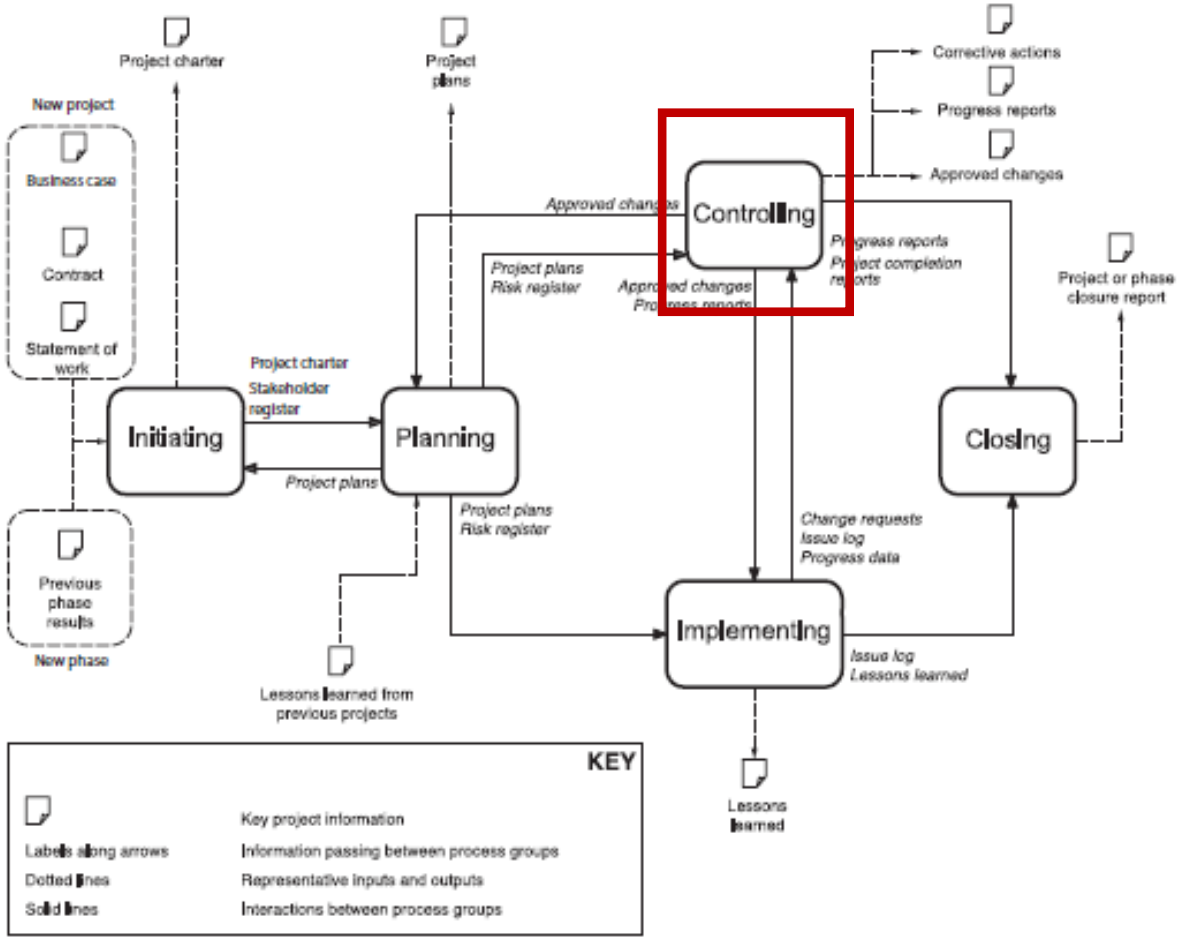
Configuration & Change Management (discussed)

Control Project Progress/Performance

Project Controlling Overview: Control Project Progress/Performance

Subject groups	Process groups				
	Initiating	Planning	Implementing	Controlling	Closing
Integration	4.3.2 Develop project charter	4.3.3 Develop project plans	4.3.4 Direct project work	4.3.5 Control project work 4.3.6 Control changes	4.3.7 Close project phase or project 4.3.8 Collect lessons learned
Stakeholder	4.3.9 Identify stakeholders		4.3.10 Manage stakeholders		
Scope		4.3.11 Define scope 4.3.12 Create work breakdown structure 4.3.13 Define activities		4.3.14 Control scope	
Resource	4.3.15 Establish project team	4.3.16 Estimate resources 4.3.17 Define project organization	4.3.18 Develop project team	4.3.19 Control resources 4.3.20 Manage project team	
Time		4.3.21 Sequence activities 4.3.22 Estimate activity durations 4.3.23 Develop schedule		4.3.24 Control schedule	
Cost		4.3.25 Estimate costs 4.3.26 Develop budget		4.3.27 Control costs	
Risk		4.3.28 Identify risks 4.3.29 Assess risks	4.3.30 Treat risks	4.3.31 Control risks	
Quality		4.3.32 Plan quality	4.3.33 Perform quality assurance	4.3.34 Perform quality control	
Procurement		4.3.35 Plan procurements	4.3.36 Select suppliers	4.3.37 Administer procurements	
Communication		4.3.38 Plan communications	4.3.39 Distribute information	4.3.40 Manage communications	

NOTE The purpose of this table is not to specify a chronological order for carrying out the activities. Its purpose is to map subject groups and process groups.



Control Project Performance

Control Schedule (ISO 21500)**:** The purpose of Control Schedule is to monitor schedule variances and to take appropriate actions. This process should focus on determining the current status of the project schedule, comparing it to the approved baseline schedule to determine any variance, forecasting completion dates and implementing any appropriate actions to avoid adverse schedule impacts. All changes to the baseline schedule should be managed in accordance with Change Management processes.

Control Costs (ISO 21500)**:** The purpose of Control costs is to monitor cost variances and to take appropriate actions. This process should focus on determining the present project cost status, comparing it to the baseline costs to determine any variance, forecasting projected costs at completion and implementing any appropriate preventive or corrective actions, in order to avoid adverse cost impacts. All changes to the baseline costs should be managed in accordance with Change Management processes.

Once work is started, performance data are accumulated including budgeted costs, actual costs and estimated cost at completion. In order to evaluate cost performance it is necessary to accumulate scheduling data, such as the progress of scheduled activities and the forecasted completion dates of current and future activities. Variances might arise from poor planning, unforeseen scope changes, technical problems, equipment failures or other external factors, such as supplier difficulties. Regardless of the cause, corrective actions require either a change in the cost baseline or the development of a short-term recovery plan.

Control Project Performance

Main Methods

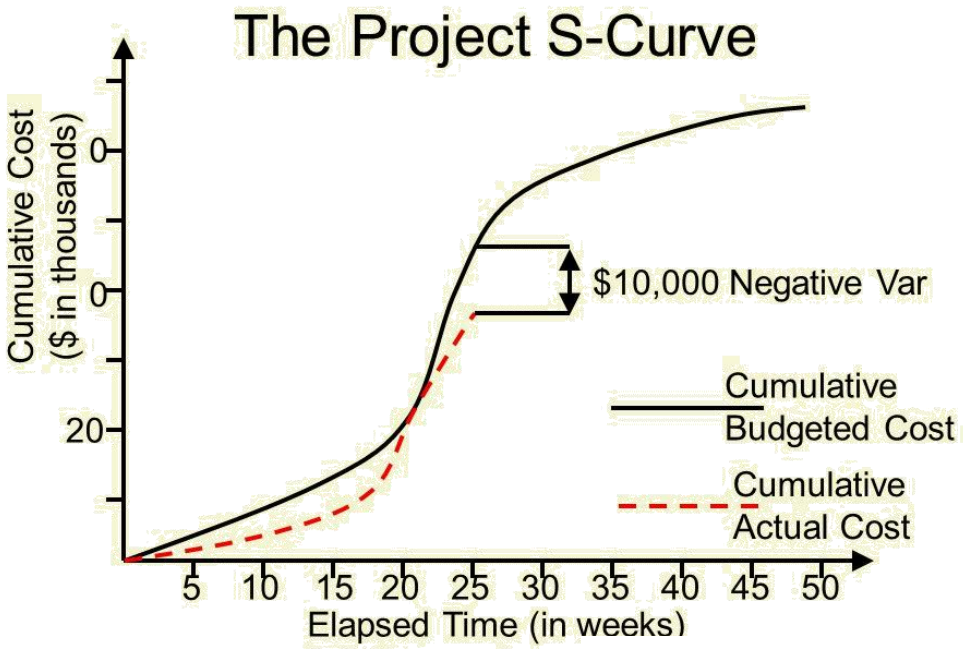
1. **S-Curve Analysis** (focus on Cost vis-a-vi Schedule)
2. **Milestones** (focus on Schedule)
3. **Tracking Gantt Charts** (focus on Schedule)
4. **Earned Value Management** (a more holistic approach)
5. **Scrum/Agile: Velocity and Sprint/Release Burndown Charts**

Control Project Performance

S-Curve Analysis (focus on Cost vis-a-vi Schedule)

Advantages: Simple, Clear, easy to Update

Main Issue: may be misinterpreted and not “tell” the whole truth on project actual performance



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Control Project Performance

Milestones (focus on Schedule):

Advantages: Simple, Clear, Widely Used.

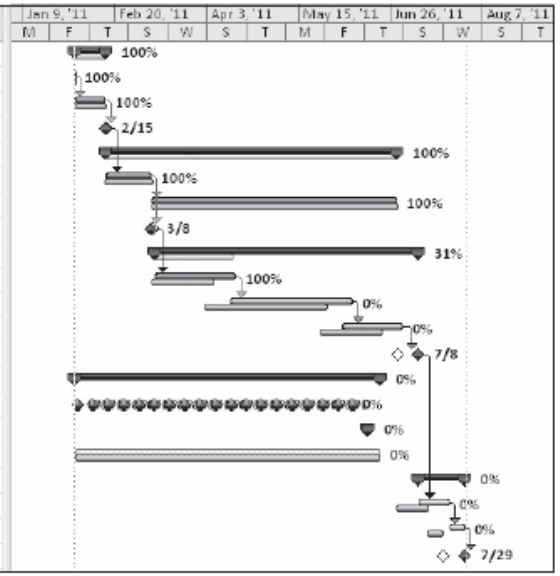
Main Issue: they are Reactive

Tracking Gantt Charts (focus on Schedule)

Advantages: Easy to understand and updated

Issues: Do not show underlying issues, do not easily allow for projections in the future

Task Name	Start	Finish	Baseline Start	Baseline Finish	Start Var.	Finish Var.
Initiating	**Tue 2/1/11**	**Tue 2/15/11**	**Tue 2/1/11**	**Tue 2/15/11**	**0 days**	**0 days**
Kickoff meeting	Tue 2/1/11	Tue 2/1/11	Tue 2/1/11	Tue 2/1/11	0 days	0 days
Develop project charter	Tue 2/1/11	Tue 2/15/11	Tue 2/1/11	Tue 2/15/11	0 days	0 days
Charter signed	Tue 2/15/11	Tue 2/15/11	Tue 2/15/11	Tue 2/15/11	0 days	0 days
Planning	**Tue 2/15/11**	**Tue 6/28/11**	**Tue 2/15/11**	**Tue 6/28/11**	**0 days**	**0 days**
Develop project plans	Tue 2/15/11	Tue 3/8/11	Tue 2/15/11	Tue 3/8/11	0 days	0 days
Review project plans	Tue 3/8/11	Tue 6/28/11	Tue 3/8/11	Tue 6/28/11	0 days	0 days
Project plans approved	Tue 3/8/11	Tue 3/8/11	Tue 3/8/11	Tue 3/8/11	0 days	0 days
Executing	**Thu 3/10/11**	**Fri 7/8/11**	**Tue 3/8/11**	**Tue 6/28/11**	**1.75 days**	**8.05 days**
Analysis	Thu 3/10/11	Fri 4/15/11	Tue 3/8/11	Tue 4/5/11	1.75 days	8.75 days
Design	Wed 4/13/11	Wed 6/8/11	Fri 4/1/11	Fri 5/27/11	8.05 days	8.05 days
Implementation	Fri 6/3/11	Fri 7/1/11	Tue 5/24/11	Tue 6/21/11	8.05 days	8.05 days
System implemented	Fri 7/8/11	Fri 7/8/11	Tue 6/28/11	Tue 6/28/11	8.05 days	8.05 days
Controlling	**Tue 2/1/11**	**Mon 6/20/11**	**Tue 2/1/11**	**Mon 6/20/11**	**0 days**	**0 days**
Status Reports	**Wed 2/2/11**	**Wed 6/8/11**	**Wed 2/2/11**	**Wed 6/15/11**	**0 days**	**-5 days**
Report performance	**Wed 6/15/11**	**Wed 6/15/11**	**Tue 2/1/11**	**Mon 6/20/11**	**96 days**	**-3.88 days**
Control changes	Tue 2/1/11	Mon 6/20/11	Tue 2/1/11	Mon 6/20/11	0 days	0 days
Closing	**Fri 7/8/11**	**Fri 7/29/11**	**Tue 6/28/11**	**Tue 7/19/11**	**8.05 days**	**8.05 days**
Prepare final project report	Fri 7/8/11	Fri 7/22/11	Tue 6/28/11	Tue 7/12/11	8.05 days	8.05 days
Present final project	Fri 7/22/11	Fri 7/29/11	Tue 7/12/11	Tue 7/19/11	8.05 days	8.05 days
Project completed	Fri 7/29/11	Fri 7/29/11	Tue 7/19/11	Tue 7/19/11	8.05 days	8.05 days



Control Project Performance

Earned Value Management (EVM)

- **PV (Planned Value):** the budget (or the baselined cost)
- **AC (Actual Cost) (cumulative)**
- **EV (Earned Value):** (estimate) of the value of the work actually completed
- **RP (Rate of Performance):** ratio of percentage of actual work completed to the percentage of work planned to have completed at any given point in time.

Example: Suppose a task is planed to take one week and a cost of £10000 (PV). Suppose that it actually took 2 weeks and a total cost of £20000, of which £15000 in the first week and £5000 in the second week (AC per week). Suppose that the task was half-way completed *by the end of week 1*. This implies that RP=50%. Calculations:

- **EV=PV (to date) * RP** = 10000*50%=5000
- **Cost Variance = EV – AC** = 5000 – 15000 = –10000
- **Schedule Variance (SV) = EV – PV** = 5000 – 10000 = – 5000, a negative value implies it took longer to complete the task
- **Cost Performance Index (CPI) = EV/AC** = 5000/15000 = 33%, if CPI <100% then over-budget, if CPI>100% then under-budget
- **Schedule Performance Index (SPI) = EV/PV** = 5000/10000 = 50%, if SPI<100% then behind schedule, if SPI>100 then ahead of schedule
- **Estimate at Completion (EAC) = Baselined Cost at Completion (BAC) / CPI**
- **Estimate Time to Complete = Original Time to Complete / SPI**

Control Project Performance

Earned Value Management (EVM)

One may produce an **earned value chart** to track project performance, i.e. example for a 1-year project after 5 months

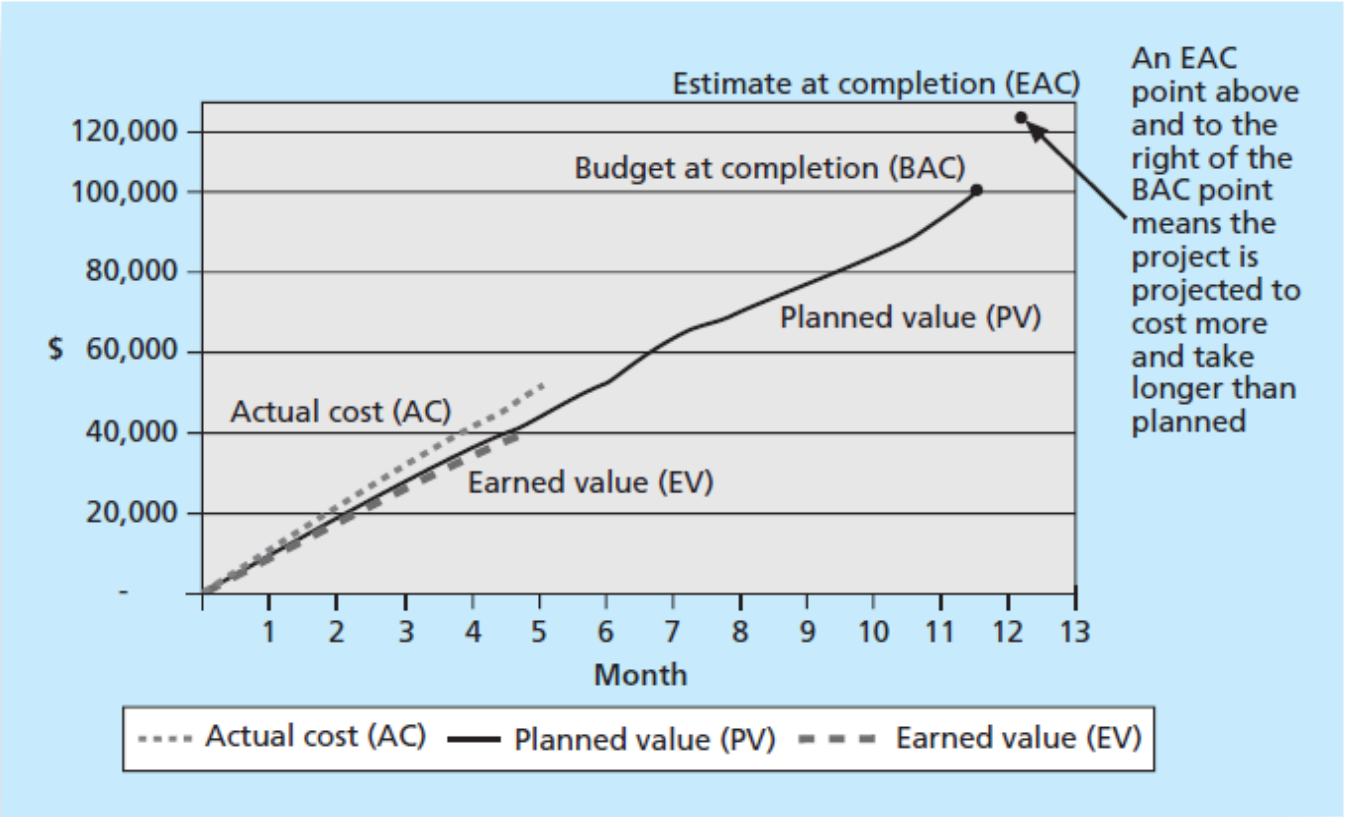


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FIGURE 7-5 Earned value chart for project after five months

Image from Schwalbe's book .

Control Project Performance

Earned Value Management (EVM): Concrete Numbers for the previous EV Graph

Earned Value Calculations for Project

Prepared by: **Date:**

Note: Change the entries to meet your project needs. This data is from Figure 7-4 of Schwalbe's text Information Technology Project Management
Also make sure the formulas work properly based on the data you enter.

Activity	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	To Date PV	Planned % Complete	Actual % Complete	RV	EV
Plan and staff project	4,000	4,000											8,000	100	100	100%	8,000
Analyze requirements		6,000	6,000										12,000	100	100	100%	12,000
Develop ERDs			4,000	4,000									8,000	100	100	100%	8,000
Design database tables				6,000	4,000								10,000	100	100	100%	10,000
Design forms, reports, and queries					8,000	4,000							8,000	75	50	67%	5,333
Construct working prototype						10,000								-			-
Test/evaluate prototype						2,000	6,000							-			-
Incorporate user feedback							4,000	6,000	4,000					-			-
Test system									4,000	4,000	2,000			-			-
Document system											3,000	1,000		-			-
Train users												4,000		-			-
Monthly Planned Value (PV)	4,000	10,000	10,000	10,000	12,000	16,000	10,000	6,000	8,000	4,000	5,000	5,000					**43,333**
Cumulative Planned Value (PV)	4,000	14,000	24,000	34,000	46,000	62,000	72,000	78,000	86,000	90,000	95,000	100,000					
Monthly Actual Cost (AC)	4,000	11,000	11,000	12,000	15,000												
Cumulative Actual Cost (AC)	4,000	15,000	26,000	38,000	53,000												
Monthly Earned Value (EV)	4,000	10,000	10,000	10,000	9,333												
Cumulative Earned Value (EV)	4,000	14,000	24,000	34,000	43,333												
Project EV as of May 31	43,333																
Project PV as of May 31	46,000																
Project AC as of May 31	$ 53,000																
CV=EV-AC	**$ (9,667)**																
SV=EV-PV	**$ (2,667)**																
CPI=EV/AC	**81.761%**																
SPI=EV/PV	**94.203%**																

Estimate at Completion (EAC) **$122,308** (original plan of $100,000 divided by CPI)

Estimated time to complete **12.74**(original plan of 12 months divided by SPI)

Control Project Performance

Scrum/Agile: Velocity and Sprint/Release Burndown Charts (additional metrics do exist)

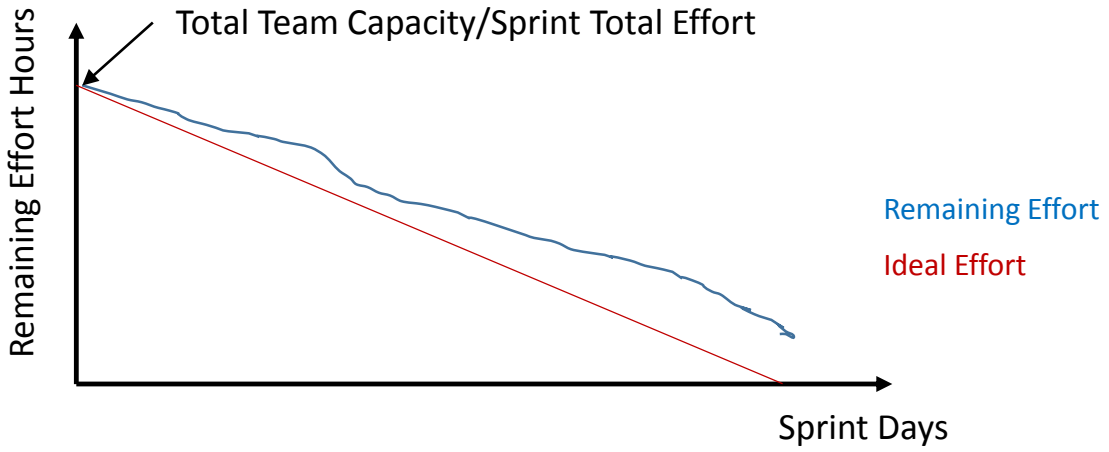
Velocity: cumulative running average of the Story Points delivered in a number of Sprints

Sprints	Planned Story Points	Actual Story Points	Velocity
Sprint 1	20	10	10
Sprint 2	30	20	15
Sprint 3	30	18	16
Sprint 4	30	20	17

Burndown Chart:

Horizontal Axis: Concrete Sprint Days

Vertical Axis: Team Capacity (Ideal Effort Hours) to Sprint Completion Vs Effort Remaining



Project Control: High-Level Summary of Best Practices

1. Decide and communicate how you are going to control change, track status and manage issues.
2. Decide *what* you will measure.
3. Decide *how* you will measure.
4. Risk Control/Management is a key success factor. Evaluate and be proactive!
5. Options available are often based on the project triangle
6. Quantify options/decisions whenever possible.
7. Recommend and Engage stakeholders Vs Decide